Stochastic Simulation - Assignment B
Ethereum Gas Market Simulation

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1 Introduction

In this report, we simulate the use of CT scanners in the radiology department. The department has three types of patients: emergency patients, inpatients and outpatients. We analyze it as a stochastic system, since patient arrivals and scan durations are random. Discrete-event simulation is used to model the problem, because relevant changes in the system occur at discrete event times, such as patient requests, patient arrivals, scan starts and scan completions.

First, the report describes the system with CT scanners and analyzes the performance indicators. After that, we analytically examine the capacity of the scanners. Subsequently, the discrete-event simulation and its statistical analysis are demonstrated. Finally, the simulation results are presented. This assignment aims to evaluate the current system's capacity and inspect performance indicators such as CT scanner utilization, waiting time, and access time.

2 Problem Description in Words

The radiology department serves three types of patient, each with a different arrival rate and service rules: emergency patients, inpatients and outpatients. There are two CT scanners available on weekdays 8:00 to 16:00, and one that works the rest of the hours. This report only considers fixed current policy. The scan duration is uniformly distributed betwen 10 and 19.

Emergency patients directly arrive at the departments, and are served as soon as possible and their arrival is modelled by Poisson process with rate $(\lambda_E = 24) per day$.

Inpatients request a scan before arrival. Requests follow Poisson distribution with rates 21 and 6 requests during office hours and outside office hours respectively. They wait until they get a call and after that they are trasported to CT with transport time being uniformly distributed between 9 and 15 minutes.

Outpatients request an appointment during office hours according to a Poisson process with rate ($\lambda_O =$ 23) per weekday. The earliest an outpatient can be scheduled is the next day. Four slots per hour are offered between 8:00 and 12:00, and three slots per hour between 12:00 and 16:00. If there are no slots available in the same week, the outpatient will be placed on a waiting list and cared for on Friday night of the next week. The probability of an outpatient to show up is $p_s = 0.84$.

Emergency patients are always served first. If no emergency patient is present, inpatients and outpatients are served in order of arrival. The waiting room has three chairs. If more than three patients are waiting, the remaining people are considered to be waiting outside the waiting room.

This assignment's performance indicators are CT scanner utilization during and outside office hours, outpatient access time, waiting times for emergency patients and outpatients, the fraction of CT patients waiting outside the waiting room, and the percentage of office-hour inpatient requests not scanned before 16:00 on the same weekday.

3 Analytical Capacity Check

We want to find out if two CT scanners are sufficient for treating all patients. This is checked analytically: the expected CT scanner workload is compared to scanner capacity.

For all patient types, the CT scan duration follows $U(10, 19)$. Let S denote the scan duration. The expected scan duration is

$$E[S] = \frac{10 + 19}{2} = 14.5 \text{ minutes} \approx 0.2417 \text{ hours}$$

Let N_E, N_I and N_O denote the number of expected weekly scans of emergency patents, inpatient and outpatient. Let W_E, W_I and W_O be their expected weekly workloads for a week.

Emergency patients arrival is a Poisson process with rate $\lambda_E = 24$ per day and $N_E = 24 \cdot 7 = 168$.

The expected emergency workload is

$$W_E = N_E E[S] = 168 \cdot 0.2417 \approx 40.60$$

hours of workload. Inpatients have on average 21 requests during office hours and 6 requests arrive outside office hours. We assume that there are 0 requests during weekends. Therefore,

$$N_I = 5(21 + 6) = 135,$$

and

$$W_I = N_I E[S] = 135 \cdot 0.2417 \approx 32.63$$

Outpatients requests have a rate of $\lambda_O = 23$ per weekday with propapability to show up $p_s = 0.84$. Therefore,

$$N_O = 5 \cdot 23 \cdot 0.84 = 96.6,$$

and the expected hours of workload:

$$96.6 \cdot 0.2417 \approx 23.35$$

The total expected scanner workload W is

$$W = W_E + W_I + W_O = 40.60 + 32.63 + 23.35 = 96.58$$

Let C_{office} and C_{outside} denote scanner capacity during office and outside office hours respectively. Two scanners are available from 8:00 to 16:00 on weekdays, so

$$C_{\text{office}} = 5 \cdot 8 \cdot 2 = 80$$

hours per week. Outside office hours, one scanner is available. There are 168 hours in a week and 40 of them are office hourse. Therefore,

$$C_{\text{outside}} = 168 - 40 = 128$$

hours per week. Hence the total capacity C:

$$C = C_{\text{office}} + C_{\text{outside}} = 80 + 128 = 208$$

Since $W < C$, we can conclude that the department's workload is smaller than its capacity.

4 Simulation Model

The CT department is modelled as a discrete-event simulation.

We make the following assumptions. Emergency patients have priority over all the other patients, while inpatients and outpatients are served in the order of arrival. A scan that has started before 16:00 is allowed to finish after 16:00. Scan duration follows $U(10, 19)$, i.e., the scan duration is distributed uniformly between 10 and 19 minutes. We ignore PAS patients

The state variables are the emergency queue, simulation time, the availability of scanners, the queue for inpatients and outpatients, the outpatient waiting list, and the outpatient appointment schedule, the number of patients who wait outside the waiting room, office-hour inpatient requests, and office-hour inpatients unscanned before 16:00 the same weekday.

The event types used the model are emergency arrival, outpatient appointment arrival, inpatient arrival, scan completion, Friday waiting-list scheduling, and capacity change.

At an emergency arrival event, a patient goes directly to the CT department: a patient is added to the emergency queue, and the waiting room counter is increased by one. If the three chairs in the waiting room are already taken, the patient is counted as waiting outside.

At an inpatient request event, the nursing ward sends a request, if inpatient is not the CT department yet. In case there are no other inpatients is waiting or being transported, inpatient is called from ward. After that, the arrival is modelled through transport time, which is uniformly distributed between 9 and 15 minutes. In case another inpatient is waiting or being transported, the request is pending.

At an inpatient arrival event, patient arrives to the department after a call from the ward, and is added to the regular queue. If the request was during office hours, the system tracks it and checks later the scan started before 16:00 on the same day. Subsequently, the waiting-room counters are renewed, and the model checks if a new scan can start.

At an outpatient request event, a patient makes an appointment. The model looks for the earliest open slot on the same week, but no earlier than the next day. If the slot is found, the appointment is planned and access time is stored. If not, the outpatient is added to the waiting list. When an outpatient arrives at the schedule time, he is added to the regular queue. In case the patient does not show up (with probability $1 - p_s$), he is not added back to the waiting list.

Every Friday evening the Friday waiting-list scheduling. The outpatients from the waiting list are assigned to all vacant appointment slots on the next week until there are no available slots.

Capacity change event happens when scanner availability changes (at 8:00 and 16:00 on weekdays). The utilization statistics is renewed, the model checks if a new scan can start.

Scan completion means that a scan is finished and the scanner is free.

5 Flowchart

6 Results

6.1 Simulation Results

The simulation results are:

Table 1: Simulation results (500 replications)

Metric	Mean	95% CI	Half-width	Rel. prec.
Office util.	0.4972	[0.4962, 0.4983]	1.07×10^{-3}	2.14×10^{-3}
Outside util.	0.3141	[0.3134, 0.3147]	6.78×10^{-4}	2.16×10^{-3}
Emerg. wait (min)	2.628	[2.611, 2.646]	1.73×10^{-2}	6.58×10^{-3}
Outpat. wait (min)	0.6737	[0.6621, 0.6854]	1.17×10^{-2}	1.73×10^{-2}
Outpat. access (days)	1.3907	[1.3877, 1.3936]	2.92×10^{-3}	2.10×10^{-3}
Frac. waited outside	0.001108	[0.001020, 0.001196]	8.77×10^{-5}	7.92×10^{-2}
Frac. missed same-day	0.000533	[0.000352, 0.000715]	1.81×10^{-4}	3.40×10^{-1}

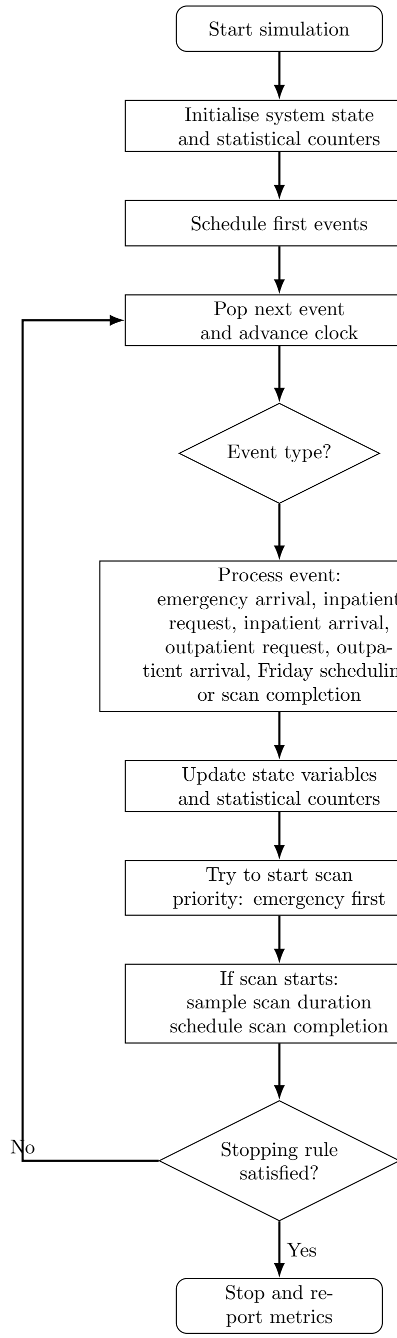

Figure 1: Flowchart of the CT scan department simulation. Events are processed in chronological order. After every event, the state and statistical counters are updated, and the model checks whether a new scan can start.

The simulation results suggest that the CT department has sufficient capacity. This matches the conclusion from the analytical capacity check. In the analytical check, the workload ratio was $\rho \approx 0.464$. Meanwhile, the model illustrated the ratios of 0.497 and 0.314 in office and outside office hours, with 95% confidence intervals $[0.496, 0.498]$ and $[0.313, 0.315]$.

Also, the results for waiting time show that the capacity is not heavily loaded. The average waiting time for emergency patients is 2.63 minutes with 95% confidence interval $[2.61, 2.65]$, and for outpatients it is 0.674 minutes with 95% confidence interval $[0.662, 0.685]$ after they arrive at the CT department. Moreover, only

0.111% of CT patients have to wait outside the waiting room, with 95% confidence interval $[0.102\%, 0.120\%]$, so the three-chair waiting room is almost always enough in this simulation run.

The average access time for outpatient visits is relatively high, about 1.39 days with 95% confidence interval $[1.388, 1.394]$. Hence, an outpatient waits more than one day on average for his appointment.

The fraction of office-hour inpatients missing the same-day target is very low, about 0.0533% with 95% confidence interval $[0.0352\%, 0.0715\%]$. Therefore, almost all office-hour inpatient requests are scanned before 16:00 on the same weekday.

7 Conclusion

To conclude, even though the time a patient waits in the CT department upon arrival is relatively low, the outpatient has to wait more than one day for his visit. As a result, the main delay in the simulation is the appointment schedule for outpatients.